

Second-order time-invariant systems

Learning goals

After this set of video lectures you will...

- Know how to construct phase portraits and interpret them
 - Analytical method
 - Computer simulations
- Be able to do a local analysis, i.e. describe the system behaviour near an equilibrium point
- Be able to describe a periodic orbit and a limit cycle
- Be able to tell whether a periodic orbit may or may not exist for a 2D system

The video lectures are based on

Chapter 2, Sections 2.1 - 2.6

Second-order time-invariant systems: Introduction

Conditions

- $n = 2$
- time-invariant (autonomous)

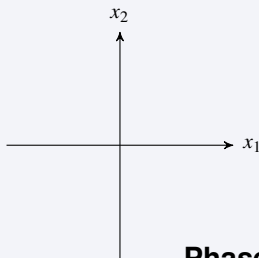
i.e.

$$\dot{x}_1 = f_1(x_1, x_2)$$

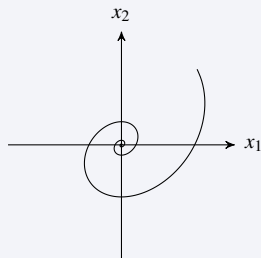
$$\dot{x}_2 = f_2(x_1, x_2)$$

Second-order time-invariant systems: Concepts

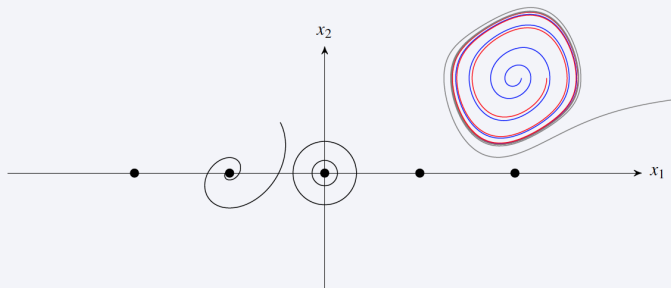
Phase plane



Trajectory



Phase portrait



Second-order time-invariant systems: Concepts

Phase plane analysis

Determine the system behavior by constructing a phase portrait

- **Singular point**

x^* :

$$\begin{bmatrix} f_1(x_1^*, x_2^*) \\ f_2(x_1^*, x_2^*) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

- The slope/tangent of a trajectory

$$\frac{dx_2}{dx_1} = \frac{\frac{dx_2}{dt}}{\frac{dx_1}{dt}} = \frac{f_2}{f_1}$$

How to construct phase portraits

How do we construct phase portraits?

- Analytical method
- Computer simulations

Analytical method - Example

Pendulum without friction

$$ml\ddot{\theta} = -mg \sin \theta$$

Math brush up

www.khanacademy.org:
Separable differential equations

Analytical method - Pendulum without friction

The trajectories are described by

$$\frac{1}{2}x_2^2 = \frac{g}{l} \cos x_1 + c$$

Computer simulations

Computer simulations

- 1 Select a region of interest in the phase plane
(Typically a bounding box
 $x_{1,\min} \leq x_1 \leq x_{1,\max}, \quad x_{2,\min} \leq x_2 \leq x_{2,\max}$)
- 2 Select a number of initial state values inside this region
- 3 Calculate and plot the corresponding trajectories, i.e. the solutions of the initial value problems

NB

For unstable equilibrium points and limit cycles, the only way to obtain a phase portrait close to these is to simulate backwards in time, i.e. to solve the IVP

$$\dot{x} = -f(x), \quad x(0) = x_0$$

Local analysis: Qualitative behavior near equilibrium points

- 1 Linearize about the equilibrium point
- 2 Find the eigenvalues $\lambda(A)$
- 3 Classify the equilibrium point

Find the equilibrium points

Equilibrium point

x^* is an equilibrium point of $\dot{x} = f(x)$ iff

$$f(x^*) \equiv 0$$

1) Linearize about the equilibrium point

Given the system

$$\dot{x}_1 = f_1(x_1, x_2)$$

$$\dot{x}_2 = f_2(x_1, x_2)$$

Taylor's formula yields

$$f_i(x_1, x_2) = \underbrace{f_i(x_1^*, x_2^*)}_{=0} + \frac{\partial f_i}{\partial x_1}(x_1^*, x_2^*)\Delta x_1 + \frac{\partial f_i}{\partial x_2}(x_1^*, x_2^*)\Delta x_2$$

$$+ \underbrace{P(\Delta x_1, \Delta x_2)}_{\text{Higher order terms} \approx 0}, \quad \Delta \mathbf{x} = \begin{bmatrix} x_1 - x_1^* \\ x_2 - x_2^* \end{bmatrix}$$

$\Downarrow$

$$\Delta \dot{\mathbf{x}} = \underbrace{\begin{bmatrix} \frac{\partial f_1}{\partial x_1}(x_1^*, x_2^*) & \frac{\partial f_1}{\partial x_2}(x_1^*, x_2^*) \\ \frac{\partial f_2}{\partial x_1}(x_1^*, x_2^*) & \frac{\partial f_2}{\partial x_2}(x_1^*, x_2^*) \end{bmatrix}}_A \Delta \mathbf{x}$$

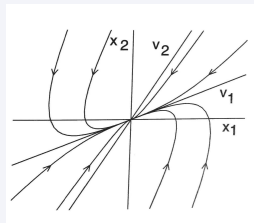
2) Find the eigenvalues $\lambda(A)$

Solve $\det(\lambda I - A) = 0$

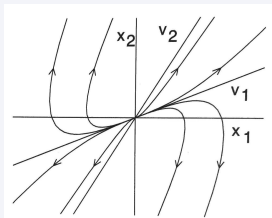
3) Classify the equilibrium points

- **Real λ**

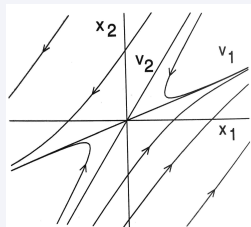
$\lambda_2 < \lambda_1 < 0$
Stable node



$0 < \lambda_1 < \lambda_2$
Unstable node



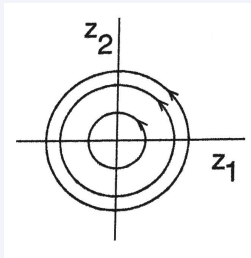
$\lambda_2 < 0 < \lambda_1$
Saddle point



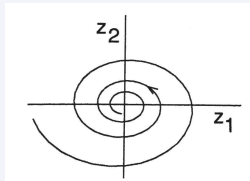
Classify the equilibrium points

- **Complex**, $\lambda_{1,2} = \alpha \pm \beta i$

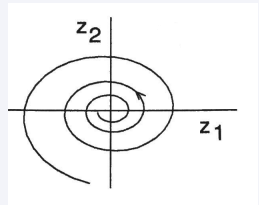
$\alpha = 0$
Center



$\alpha < 0$
Stable focus



$\alpha > 0$
Unstable focus



Why perform a local analysis?

Nonlinear second-order system

$$\dot{x}_1 = f_1(x_1, x_2)$$

$$\dot{x}_2 = f_2(x_1, x_2)$$

where $f_1 : \mathbb{D} \rightarrow \mathbb{R}^n$ and $f_2 : \mathbb{D} \rightarrow \mathbb{R}^n$ are C^1 functions,
and $\mathbb{D} \subset \mathbb{R}^n$ is a neighborhood of the equilibrium point.

Topological equivalence

If $\text{Re}(\lambda_{1,2}) \neq 0$ then

the local
phase portrait



the phase portrait of
the linearized system

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Nonlinear second-order system

$$\dot{x}_1 = f_1(x_1, x_2)$$

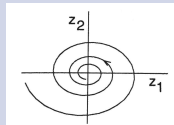
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Topological equivalence

If $\text{Re}(\lambda_{1,2}) \neq 0$ then

the local
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How to do a phase plane analysis

Summary: How to do a phase plane analysis

- ➊ Find the equilibrium points of the system
- ➋ Perform a local analysis (about each equilibrium point)
 - ➊ Linearize about the equilibrium point
 - ➋ Find the eigenvalues $\lambda(A)$
 - ➌ Classify the equilibrium point

In order to obtain qualitative knowledge about the system behavior locally around the equilibria. (This will guide you in the next step)
- ➌ Construct a phase portrait using
 - a) the analytical method
 - b) computer simulations
- ➍ Try to find possible periodic orbits and limit cycles

Periodic Orbits and Limit cycles

Definition: Limit cycle

A non-trivial isolated periodic orbit.

Periodic orbit

$\exists T > 0$ such that

$$x(t+T) = x(t) \quad \forall t \geq 0$$

NB Linear systems do not have limit cycles

Example

- Temperature oscillations in a room controlled by a relay with hysteresis.
- Course of a ship having backlash in its rudder machine
- Electronic oscillator

Periodic orbits and Limit cycles: Existence criteria

Existence criteria for periodic orbits

1. Poincaré-Bendixson
2. The Bendixson (negative) criterion
3. The index method

Poincaré-Bendixson criterion

Lemma 2.1 (Poincaré-Bendixson criterion)

Consider the system

$$\dot{x} = f(x), \quad f : \mathbb{R}^2 \rightarrow \mathbb{R}^2 \text{ is } C^1$$

Let M be a closed bounded subset of the plane such that...

Poincaré-Bendixson criterion

Lemma 2.1 (Poincaré-Bendixson criterion)

Consider the system

$$\dot{x} = f(x), \quad f : \mathbb{R}^2 \rightarrow \mathbb{R}^2 \text{ is } C^1$$

Let M be a closed bounded subset of the plane such that.

- M contains no equilibrium points of the system, or it contains only one equilibrium point with the property that the eigenvalues of the Jacobian matrix at this point have positive real parts (unstable focus or unstable node)
- Every trajectory starting in M stays in M for all future time

Then M contains a periodic orbit of the system.

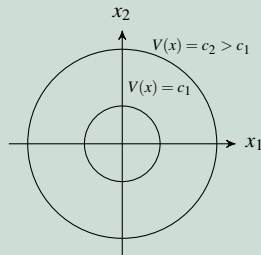
Poincaré-Bendixson criterion: Example

Example 2.7

Consider the harmonic oscillator, given by

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= -x_1\end{aligned} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} x, \quad \lambda = \pm j$$

1) $V(x) = c$

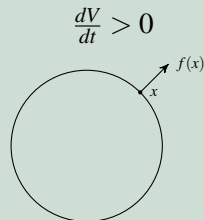
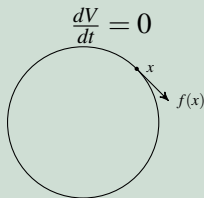
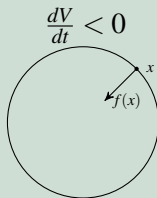


Poincaré-Bendixson criterion: Example cont.

Example 2.7 cont.

2) Calculate the directional derivative

$$\frac{dV}{dt} = \frac{\partial V}{\partial x} \cdot \frac{\partial x}{\partial t} = \nabla V \cdot f|_{x \in \{x \in \mathbb{R}^2 : V(x) = c\}}$$



Poincaré-Bendixson criterion: Example cont.

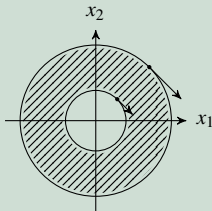
Example 2.7 cont.

1) $V(x) = x_1^2 + x_2^2$

2) $\nabla V \cdot f|_{V(x)=c} = \frac{\partial V}{\partial x} \cdot f|_{V(x)=c} = [2x_1 \quad 2x_2] \begin{bmatrix} x_2 \\ -x_1 \end{bmatrix} = 0$

- Trajectories do not leave $\{x \in \mathbb{R}^2 : V(x) \leq c\}$
- But: Contains an equilibrium point - not unstable focus/node
- Solution: Exclude the region where the equilibrium point is

$$M = \{x \in \mathbb{R}^2 : c_1 \leq V(x) \leq c_2\}$$



Trajectories do not leave M

$\Rightarrow$ There is a periodic orbit in M

Note: Ensures existence, not uniqueness

Bendixson (negative) Criterion

Lemma 2.2 Bendixson (negative) Criterion

Consider the system

$$\dot{x} = f(x), \quad x \in \mathbb{R}^2.$$

If, on a simply connected region D of the plane, the expression

$$\frac{\partial f_1}{\partial x_1} + \frac{\partial f_2}{\partial x_2}$$

is not identically zero and does not change sign, then the system has no periodic orbits lying entirely in D .

The index theorem

Corollary 2.1 The index theorem

C is a periodic orbit $\Rightarrow \sum_i I(\text{eq. point } i) \text{ within } C = +1$

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$$I(\text{node, focus, center}) = +1$$

$$I(\text{saddle point}) = -1$$